

Tran Quoc Truong

Applied AI Engineer · AI Agents & RAG · Backend Systems

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Summary

Applied AI Engineer focused on translating business and operational challenges into practical, reliable AI solutions. Combines solution architecture and hands-on engineering to address enterprise knowledge access, intelligent automation, and AI agent reliability, with end-to-end ownership from problem framing and architectural decisions to delivery and quality validation.

Technical Skills

AI & Agentic Systems: RAG, Hybrid Retrieval, LangGraph, MCP (Model Context Protocol), Tool Calling, LiteLLM, Qdrant, Docling

ML, Fine-Tuning & Model Serving: PyTorch, Hugging Face Transformers, PEFT, Unsloth, LoRA/QLoRA, vLLM, Synthetic Data Generation

LLM Evaluation & Reliability: OpenTelemetry, Runtime Trace Reconstruction, LLM-as-a-Judge, Quality Gates, Guardrails, Langfuse

Backend Engineering: Python, FastAPI, Django/DRF, Pydantic, SQLAlchemy, PostgreSQL, Redis, Celery, SSE / Streaming APIs

Test Automation & Quality Engineering: Playwright, Cypress, Katalon Studio, Postman, API Testing, Regression Testing, SQL Validation

Infrastructure & Delivery: Docker, Docker Compose, RunPod, Nginx, MinIO, CI/CD, Linux

Experience

FPT Software , Applied AI Engineer & Automation Tester (IVS)	Ho Chi Minh City, Vietnam
IQP (Code Intelligence): Designed multi-repository code intelligence combining semantic, lexical, and code-graph retrieval to provide traceable implementation and test context through MCP-enabled AI workflows. Agent Assurance: Architected trace-based evaluation that reconstructs agent execution evidence against versioned requirements, supporting auditable regression analysis and governed quality-gate decisions. Banking Automation: Reduced manual regression effort by ~70% across banking and payment workflows using reusable API, database, and cross-platform automation with SQL-based transaction validation.	Dec 2024 – present 1 year 11 months
FPT Software , Full-Stack Developer Intern (FSA)	Ho Chi Minh City, Vietnam
- Built the core backend for an online learning and assessment platform using Django with role-based access control and automated exam workflows. - Integrated a separate FastAPI computer-vision service for AI-assisted real-time exam proctoring and face detection. - Containerized and deployed the multi-service architecture using Docker Compose, Nginx reverse proxy, and Cloudflare Tunnel.	Sept 2024 – Dec 2024 4 months

Achievements

Golden Solution Prize (1st Place) — IVS Solution Day 2.0 2026	Sept 2026
Golden Solution Prize at FPT Software’s IVS Solution Day for Flezi Polaris (Agent Assurance), focused on evidence-backed AI evaluation and quality verification.	
2nd Runner-up (3rd Place) — IVS Hackathon 2026	Aug 2026
2nd Runner-up (3rd place) at FPT Software’s IVS Hackathon for independently architecting the Agent Assurance engine later integrated into Omni-Agent.	
Second Prize — Autonomous Driving Research Paper Competition	Apr 2024
Second Prize for sole-authored autonomous-driving research combining YOLOv8 perception, lane segmentation, and PID control.	

Featured Projects

Omni-Agent — Applied AI Platform (Knowledge Builder & Agent Assurance)

Independently architected and built a pre-production Applied AI platform for enterprise document intelligence, grounded multi-agent workflows, and reproducible AI-agent assurance.

- **Solution Architecture:** Designed project-scoped FastAPI services with independent persistence, authorization boundaries, and durable asynchronous processing across knowledge and assurance workloads.
- **Knowledge Engineering:** Built versioned Docling/OCR processing and PostgreSQL–Qdrant hybrid retrieval, preserving usable documents through recoverable indexing failures.
- **Agentic RAG & Trust:** Orchestrated LangGraph research agents through MCP, with run-scoped citations, immutable answer evidence, and input/context/output guardrails.
- **Agent Assurance:** Designed versioned executable Scenarios, qualified Run snapshots, OpenTelemetry evidence capture, and deterministic quality gates for repeatable AI-agent evaluation.
- **Product Engineering:** Streamlined first-test onboarding into Connect → Define Test → Run & Result, reducing required actions from 33 to 6 for a test API (4 for demo) in a controlled Playwright fixture while preserving preflight and informed consent.

APIT — Agent Programmatic Integration Testing

AI-assisted API testing framework generating structured test scenarios from API documentation via fine-tuned LLMs.

- Developed a bilingual synthetic dataset generation pipeline producing 1,258 domain-specific API testing samples.
- Fine-tuned Qwen2.5-3B with LoRA/QLoRA, achieving macro-F1 of 0.655 (+41.5% relative improvement over Llama-3.2-3B baseline).
- Deployed the fine-tuned model on RunPod through an OpenAI-compatible vLLM service with quantized inference and runtime LoRA adapters.
- Implemented a hybrid evaluation pipeline combining LLM-as-a-Judge, fuzzy matching, and deterministic JSON-schema validation.

E-Commerce AI Assistant — Conversational RAG & Search

End-to-end shopping assistant combining filtered vector search and multi-agent conversational retrieval.

- Built filtered semantic product retrieval with Milvus and conversational consultation through LangGraph/LangChain.
- Delivered a Docker-based prototype integrating FastAPI, PostgreSQL, Redis, and vector search.

Education

FPT School of Business & Technology , Master of Software Engineering in AI	Ho Chi Minh City, Vietnam May 2026 – present
FPT University , Bachelor of Artificial Intelligence	Ho Chi Minh City, Vietnam 2021 – 2025

Languages

- Vietnamese:** Native
- English:** Professional working proficiency